# Appendix to *Quality-Aware Data Selection for Downstream Models over Various Data Types*

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This appendix accompanies the submitted manuscript and the code repository. Section 1 restates the definitions the proofs use. Section 2 proves every formal result of the manuscript with the steps written out. Section 3 states four results that the manuscript does not contain. Section 4 records the method and protocol details that the manuscript leaves to this appendix, and Section 5 gives further measurements behind its findings. Section 6 documents, for each published method that the benchmark re-implements, what the implementation retains and where it departs from the published formulation. Section 7 lists the source and the derived version of every dataset. Numbering of the manuscript is cited as "Theorem 1 of the paper" and so on, while results of this appendix carry their own numbers.

## 1 Notation and premises

A task supplies a candidate pool $\mathcal{D} = \{x_i\}_{i=1}^{N}$ of unified records, a per-record cost $c(x) > 0$, and a budget $B$. A selection strategy $f$ returns $f(\mathcal{D}) \subseteq \mathcal{D}$ with $c(f(\mathcal{D})) \leq B$, and $k$ denotes the number of selected records (Definition 3.1 of the paper). The task also fixes a downstream model, a training or conditioning procedure, and a metric oriented so that larger is better. The test utility $g(f)$ is the expected metric of the model produced from $f(\mathcal{D})$, and the validation utility $\widehat{u}_{\mathcal{V}}(f)$ is the metric of the same model on a validation sample $\mathcal{V}$ (Definition 3.2 of the paper). A strategy is score-only when its output distribution depends on pool observables and validation records alone and on no output of the downstream model trained or conditioned on a candidate subset (Definition 3.3 of the paper).

A portfolio $\mathcal{F} = \{f_1, \ldots, f_m\}$ is a finite list of strategies fixed before the ranking split is read. Its reference-eligible subset $\mathcal{R}$ contains the strategies that read no validation data or only the construction split, and it always contains uniform random selection. The reference $f_{\mathrm{ref}}$ maximizes the ranking-split utility over $\mathcal{R}$, the challenger $f_{\mathrm{ch}}$ maximizes it over $\mathcal{F}$, and the elected strategy $f^\star \in \{f_{\mathrm{ref}}, f_{\mathrm{ch}}\}$ is the output of the adoption rule (Definition 3.4 of the paper). The labeled validation data are split into the construction split $\mathcal{V}_{\mathrm{con}}$, the ranking split $\mathcal{V}_{\mathrm{rank}}$ with $n = n_{\mathrm{rank}}$ units, and the confirmation split $\mathcal{V}_{\mathrm{conf}}$.

The error bounds use four premises (Definition 4.2 of the paper). (P1) The metric is a mean of per-unit scores in $[0, 1]$, or a two-sample U-statistic with a kernel in $[0, 1]$ such as ROC AUC, in which case $n$ denotes $\min(n_+, n_-)$. (P2) Ranking units and test units are independent draws from one distribution, or, for scalar per-unit metrics, simple random samples without replacement from one fixed finite population. (P3) Every candidate, every screening decision, and every synthesized

challenger is a function of the pool and of $\mathcal{V}_{\text{con}}$ only, and $\mathcal{V}_{\text{con}}$ is independent of $\mathcal{V}_{\text{rank}}$. (P4) The model that the ranking split scores is the model whose test metric is reported.

# 2 Detailed proofs

## 2.1 Cooperative purity and coverage

Proposition 1 of the paper bounds the contamination that survives an admission step and the coverage that an inner selector keeps. The statement is restated here with the proof written out.

**Proposition 2.1** (Proposition 1 of the paper)**.** *Fix a representation with pool density $p = \pi p_0 + (1-\pi)p_1$ for some $0 < \pi < 1$, where $p_0$ is the density of clean records and $p_1$ the density of dirty records, and let the construction split have density $p_0$. The Bayes discriminator between the two sources under equal source priors is $d = \frac{p_0}{p_0+p}$, and the clean posterior $\varphi = \frac{\pi d}{1-d}$ has the same ranking as $d$. Let the pool follow this mixture independently across records, let $Z_i$ indicate cleanliness, and let $\mathcal{I}$ contain the observed features, the construction information that is independent of the pool, and independent algorithmic randomness, so that $\mathbb{E}[Z_i \mid \mathcal{I}] = \varphi(x_i)$. Let the fitted scores, the admission, and the selection be $\mathcal{I}$-measurable. Work on the event $\max_i |\widehat{d}_i - d(x_i)| \le e$. Let $A$ be the finite admissible set after the filters and the refill, with $1 \le k \le |A|$ and $\widehat{d}_i \ge t$ for every $x_i \in A$, where $0 \le t - e < 1$. Then every selection $S \subseteq A$ of $k$ records satisfies*

$$\mathbb{E}\left[1 - \frac{1}{k}\sum_{x_i \in S} Z_i \;\middle|\; \mathcal{I}\right] \le 1 - \min\left\{1, \frac{\pi(t-e)}{1-t+e}\right\}. \tag{1}$$

*For Euclidean farthest-first selection, let $r_k(A)$ be the optimal $k$-center radius of $A$ and let $h(T, A) = \max_{x \in T} \min_{a \in A} \|x - a\|$ for a finite nonempty target set $T$. Then*

$$\max_{x \in T} \min_{y \in S} \|x - y\| \le h(T, A) + 2r_k(A). \tag{2}$$

*Proof. The discriminator and the clean posterior.* Draw a record from the construction source and from the pool source with probability $\frac{1}{2}$ each. Bayes' rule gives

$$\Pr(\text{construction} \mid x) = \frac{\frac{1}{2}p_0(x)}{\frac{1}{2}p_0(x) + \frac{1}{2}p(x)} = \frac{p_0(x)}{p_0(x) + p(x)} = d(x),$$

hence $\frac{d}{1-d} = \frac{p_0}{p}$ and $\frac{\pi d}{1-d} = \frac{\pi p_0}{p}$. Since a pool record is clean with probability $\frac{\pi p_0(x)}{p(x)}$ given $x$, the identity $\varphi = \mathbb{E}[Z \mid x]$ holds. The map $u \mapsto \frac{\pi u}{1-u}$ is strictly increasing on $[0, 1)$, so $\varphi$ and $d$ rank records identically.

*Purity.* Because the selection is $\mathcal{I}$-measurable and $\mathbb{E}[Z_i \mid \mathcal{I}] = \varphi(x_i)$,

$$\mathbb{E}\left[\frac{1}{k}\sum_{x_i \in S} Z_i \;\middle|\; \mathcal{I}\right] = \frac{1}{k}\sum_{x_i \in S} \varphi(x_i).$$

On the event of the score error, every $x_i \in A$ satisfies $d(x_i) \ge \widehat{d}_i - e \ge t - e$, and $\varphi$ is nondecreasing in $d$, so $\varphi(x_i) \ge \frac{\pi(t-e)}{1-(t-e)}$. A probability never exceeds one, which gives the minimum with one and then (1).

*Coverage.* Let $S = \{y_1, \ldots, y_k\}$ be the farthest-first centers of $A$ and let $R = \max_{x \in A} \min_j \|x - y_j\|$ be the radius they achieve. Let $x^*$ attain $R$. Farthest-first traversal picks each center at the largest

distance to the centers chosen before it, so the $k+1$ points $y_1, \ldots, y_k, x^*$ are pairwise at distance at least $R$. Any optimal $k$-center solution covers $A$ by $k$ balls of radius $r_k(A)$, so two of these $k+1$ points lie in one ball, and the triangle inequality gives $R \le 2r_k(A)$. For $x \in T$ choose $a \in A$ with $\|x-a\| \le h(T,A)$ and $y \in S$ with $\|a-y\| \le R$, which yields (2). □

The two bounds move in opposite directions in the admission width $\rho$. A narrower admission raises $t$ and tightens (1), and it can enlarge $h(T,A)$ in (2), since the admitted set it leaves to the filters is smaller.

### 2.2 Identification limit of score-only selection

**Theorem 2.2** (Theorem 1 of the paper)**.** *For every $\gamma \in (0,1]$ there exist two selection environments with identical pool observables, each admitting a selection with zero excess risk, such that every score-only strategy incurs expected excess risk at least $\frac{\gamma}{2}$ in one of them. A score-only strategy attains exactly $\frac{\gamma}{2}$ in both environments if and only if its expected allocation to the two indistinguishable groups is balanced, and uniform random selection is one such strategy.*

*Proof.* Take a pool of $2n_0$ records, $n_0$ of them carrying an observable flag $a=1$ and $n_0$ carrying $a=0$, with every other observable identically distributed across the two halves, and let the budget admit $n_0$ records. The two environments share the pool, the observables, and the validation records, and they differ only in the learner that consumes the selected records. For a selected set $S$ write $\alpha(S)$ for the fraction of $S$ with $a=0$, with the convention $\alpha(\emptyset)=\frac{1}{2}$.

In environment $E_1$ the records with $a=0$ are harmful and the excess risk of $S$ is $\gamma\,\alpha(S)$, which is zero for $S=\{a=1\}$. In environment $E_2$ the same records are the boundary records that the learner needs and the excess risk is $\gamma\,(1-\alpha(S))$, which is zero for $S=\{a=0\}$. Each environment therefore admits a selection of zero excess risk.

A score-only strategy reads pool observables and validation records only, and these agree in the two environments, so its output distribution is the same in both. Write $\bar{\alpha} = \mathbb{E}[\alpha(S)]$ for that single distribution. Its expected excess risks are $\gamma\bar{\alpha}$ in $E_1$ and $\gamma(1-\bar{\alpha})$ in $E_2$, and

$$\max\{\gamma\bar{\alpha},\, \gamma(1-\bar{\alpha})\} \ge \gamma \max\{\bar{\alpha}, 1-\bar{\alpha}\} \ge \frac{\gamma}{2},$$

with equality in both coordinates if and only if $\bar{\alpha}=\frac{1}{2}$. Uniform selection of $n_0$ of the $2n_0$ records gives $\bar{\alpha}=\frac{1}{2}$. □

A rule that maps pool features to one of several score-only strategies reads no downstream model either, so it is itself score-only and the theorem covers feature-based strategy selection. Validation utilities separate the two environments. With the flag strategy, its inversion, and uniform random selection in $\mathcal{R}$, and with validation units that report scores in $[0,1]$ whose mean is one minus the excess risk, part (a) of Theorem 2.4 with the range-based radius $2r_1$ at level $\delta_1$ bounds the expected excess risk of the elected strategy by $2r_1 + \delta_1\gamma$. This falls below $\frac{\gamma}{2}$ once $\delta_1 < \frac{1}{4}$ and

$$2\sqrt{\frac{1}{2n_{\mathrm{rank}}}\ln\frac{2m}{\delta_1}} \le \frac{\gamma}{4}, \qquad \text{that is} \qquad n_{\mathrm{rank}} \ge \frac{32}{\gamma^2}\ln\frac{2m}{\delta_1}.$$

### 2.3 Disagreement-dependent validation complexity

**Theorem 2.3** (Theorem 2 of the paper)**.** *For every $m \ge 16$, $\gamma \in (0, \frac{1}{2}]$, and $q \in [\frac{\gamma}{2}, \frac{1}{2}]$ there exist $m$ environments with identical pool observables and $m$ strategies such that $f_j$ is optimal in environment $E_j$, every other strategy has excess risk $\frac{\gamma}{2}$, and every distinct pair of strategies has per-unit squared*

*score difference of expectation $q$. Any procedure with no other environment-dependent information that reads $n$ independent shared validation units satisfies*

$$\max_{j\le m} \mathbb{E}_{E_j}[\text{excess}] \ge \frac{\gamma}{2}\left(1 - \frac{\frac{n\gamma^2}{2q} + \ln 2}{\ln m}\right). \tag{3}$$

*Consequently $n \le \frac{q(\ln m - \ln 4)}{\gamma^2}$ forces expected excess risk at least $\frac{\gamma}{4}$ in some environment.*

*Proof. Construction.* Partition the pool into $m$ observable groups of equal size, let the budget admit one group, and let $f_j$ select group $j$. A validation unit first draws a common activation bit $A \sim \text{Bernoulli}(2q)$, which is legitimate because $q \le \frac{1}{2}$. If $A = 0$, every strategy scores zero on the unit and nothing else is revealed. If $A = 1$, the unit reveals independent scores in $\{0, 1\}$, with mean $\frac{1}{2} + \frac{\gamma}{4q}$ for $f_j$ under $E_j$ and mean $\frac{1}{2}$ for every other strategy. The parameter is a probability because $q \ge \frac{\gamma}{2}$ gives $\frac{\gamma}{4q} \le \frac{1}{2}$.

*Utilities and disagreement.* The expected per-unit score is $2q\left(\frac{1}{2} + \frac{\gamma}{4q}\right) = q + \frac{\gamma}{2}$ for the optimal strategy of the environment and $2q \cdot \frac{1}{2} = q$ for every other strategy, so the utility gap is $\frac{\gamma}{2}$ and the excess risk of a suboptimal strategy is $\frac{\gamma}{2}$. For two distinct strategies the squared score difference is the indicator that the two scores differ, so its expectation is $2q \cdot \Pr(\text{two independent fair draws differ}) = 2q \cdot \frac{1}{2} = q$ when neither strategy is the optimal one, and the same value up to the mean shift of the active coordinate for a pair that contains it.

*Information.* Let $P_0$ be the auxiliary law in which every active mean equals $\frac{1}{2}$. Conditioned on $A = 1$ the laws $P_j$ and $P_0$ differ in one coordinate only, so with $\mathrm{kl}(p, r) = p \ln \frac{p}{r} + (1 - p) \ln \frac{1-p}{1-r}$ and the bound $\mathrm{kl}(p, r) \le \frac{(p-r)^2}{r(1-r)}$,

$$\mathrm{KL}(P_j \| P_0) \le 2q\,\mathrm{kl}\left(\tfrac{1}{2} + \tfrac{\gamma}{4q}, \tfrac{1}{2}\right) \le 2q \cdot \frac{\left(\frac{\gamma}{4q}\right)^2}{\frac{1}{4}} = \frac{\gamma^2}{2q}.$$

Let $J$ be uniform on $\{1, \dots, m\}$ and let $Y^n$ be the $n$ units. Mutual information is bounded by the average divergence to any fixed law, so $I(J; Y^n) \le \frac{1}{m}\sum_j \mathrm{KL}(P_j^{\otimes n} \| P_0^{\otimes n}) \le \frac{n\gamma^2}{2q}$.

*Fano step.* Read the output of the procedure as an estimate $\widehat{J}$ of the environment index. Fano's inequality gives

$$\Pr(\widehat{J} \ne J) \ge 1 - \frac{I(J; Y^n) + \ln 2}{\ln m},$$

and internal randomization is covered by averaging. A wrong index costs excess risk exactly $\frac{\gamma}{2}$, so $\mathbb{E}[\text{excess}] = \frac{\gamma}{2}\Pr(\widehat{J} \ne J)$ under the uniform prior, and the maximum over environments is at least the average, which is (3).

*Numerical form.* With $n \le \frac{q(\ln m - \ln 4)}{\gamma^2}$,

$$\frac{\frac{n\gamma^2}{2q} + \ln 2}{\ln m} \le \frac{\frac{1}{2}\ln \frac{m}{4} + \ln 2}{\ln m} = \frac{\frac{1}{2}\ln m - \ln 2 + \ln 2}{\ln m} = \frac{1}{2},$$

so the bracket in (3) is at least $\frac{1}{2}$ and the excess risk is at least $\frac{\gamma}{4}$. □

Two consequences are worth recording. Reading no validation unit recovers the score-only regime, since $n = 0$ turns the right side of (3) into $\frac{\gamma}{2}\left(1 - \frac{\ln 2}{\ln m}\right)$, which approaches the bound of Theorem 2.2 as $m$ grows. In the opposite direction, the paired radius of Theorem 2.4 with $\tau = 0$ reaches utility gap $\gamma$ at $n$ of order $\frac{q \ln m}{\gamma^2}$ at fixed confidence, so the order in (3) is attained and the family above, including the regime $q \asymp \gamma$, is the hardest one for a shared validation split.

## 2.4 Paired adjudication guarantee

**Theorem 2.4** (Theorem 3 of the paper)**.** *Condition on the pool, the construction data, and the training randomness. Let $\mathcal{F}$ contain $m \geq 2$ frozen candidates with reference set $\mathcal{R}$, write $\widehat{u}_f = \widehat{u}_{\mathrm{rank}}(f)$, and let $s_f$ be the per-unit metric score of the frozen model of $f$. For independent ranking units put $D_i^{fb} = s_f(X_i) - s_b(X_i) \in [-1, 1]$, $\widehat{q}_{fb} = \frac{1}{n}\sum_{i\leq n}(D_i^{fb})^2$,*

$$L = \ln \frac{3\binom{m}{2}}{\delta_1}, \qquad Q_{fb} = \min\left\{1,\ \widehat{q}_{fb} + \sqrt{\frac{2\widehat{q}_{fb}L}{n}} + \frac{2L}{n}\right\},$$

$$B_{fb} = \sqrt{\frac{2Q_{fb}L}{n}} + \frac{2L}{3n}, \qquad B_{ff} = 0.$$

*With probability at least $1 - \delta_1$, every pair satisfies $|\widehat{u}_f - \widehat{u}_b - g(f) + g(b)| \leq B_{fb}$. On that event the margin election satisfies*

$$\text{(a) } g(f^\star) \geq \max_{b\in\mathcal{R}}\{g(b) - B_{f^\star b}\}, \quad \text{(b) } g(f^\star) \geq \max_{b\in\mathcal{F}}\{g(b) - B_{f^\star b}\} - \tau,$$

$$\text{(c) } g(f_{\mathrm{ch}}) - g(f_{\mathrm{ref}}) > \tau_{\mathrm{sw}} - B_{f_{\mathrm{ch}}f_{\mathrm{ref}}}$$

*for every adoption whose empirical gain exceeds $\tau_{\mathrm{sw}}$, where $\tau_{\mathrm{sw}} = \eta|\widehat{u}_{f_{\mathrm{ref}}}|$ and $\tau = \widehat{u}_{f_{\mathrm{ch}}} - \widehat{u}_{f^\star}$.*

*Proof. A bound on the paired second moment.* Fix a pair and write $D$ for its per-unit difference and $q = \mathbb{E}D^2$. The variable $W = D^2$ takes values in $[0, 1]$ and satisfies $\mathrm{Var}(W) \leq \mathbb{E}W^2 \leq \mathbb{E}W = q$, because $W \leq 1$. The lower tail of Bernstein's inequality for a variable bounded in $[0, 1]$, in the form used for empirical Bernstein bounds [13], gives

$$\Pr\left(q - \widehat{q} \geq \sqrt{\frac{2qL}{n}}\right) \leq e^{-L}.$$

Outside that event, $q \leq \widehat{q} + \sqrt{\frac{2qL}{n}}$. Substituting $u = \sqrt{q}$ turns this into $u^2 - \sqrt{\frac{2L}{n}}\,u - \widehat{q} \leq 0$, so

$$\sqrt{q} \leq \frac{1}{2}\left(\sqrt{\frac{2L}{n}} + \sqrt{\frac{2L}{n} + 4\widehat{q}}\right),$$

and squaring, together with $\sqrt{xy} \leq \frac{x+y}{2}$, gives $q \leq \widehat{q} + \sqrt{\frac{2\widehat{q}L}{n}} + \frac{2L}{n}$. Since $q \leq 1$ as well, $q \leq Q_{fb}$.

*A bound on the paired mean.* The difference satisfies $\mathrm{Var}(D) \leq \mathbb{E}D^2 = q$ and $|D - \mathbb{E}D| \leq 2$, so Bernstein's inequality gives

$$\Pr\left(\left|\bar{D} - \mathbb{E}D\right| \geq \sqrt{\frac{2qL}{n}} + \frac{2L}{3n}\right) \leq 2e^{-L}.$$

Under (P3) the candidates and their frozen models are functions of the pool and $\mathcal{V}_{\mathrm{con}}$, so conditioning on them leaves the ranking units independent draws, and under (P4) the expectation $\mathbb{E}D = g(f) - g(b)$ refers to the reported models. Replacing $q$ by $Q_{fb}$ is valid on the event of the first step, so a pair fails with probability at most $3e^{-L}$. A union bound over the $\binom{m}{2}$ pairs and the choice of $L$ give failure probability at most $3\binom{m}{2}e^{-L} = \delta_1$.

*From the paired bound to the election.* The election maximizes $\widehat{u}$ over $\mathcal{R}$, hence $\widehat{u}_{f^\star} \geq \widehat{u}_b$ for $b \in \mathcal{R}$, and the paired bound turns this into (a). Over $\mathcal{F}$ the elected strategy is within the empirical

deficit $\tau$ of the best candidate, which gives (b). For (c), an adoption requires $\widehat{u}_{f_{\mathrm{ch}}} - \widehat{u}_{f_{\mathrm{ref}}} > \tau_{\mathrm{sw}}$, and subtracting the paired radius from the empirical gain gives the population statement.

*Sampling without replacement.* For scalar per-unit scores sampled without replacement from a fixed finite population, the sum of the sampled paired differences is dominated in convex order by the corresponding sum with replacement [1], so both exponential-moment bounds above hold unchanged, with $g$ and $q$ defined on that population after conditioning on the construction data.

*Two-sample U-statistics.* For ROC AUC the paired statistic is an average over all pairs of a positive and a negative unit. Split the set of such pairs into $\max(n_+, n_-)$ matchings, each of which consists of $\min(n_+, n_-)$ independent disjoint pairs. The statistic is the average of the matching means, each of which obeys the bounds above with $n = \min(n_+, n_-)$ independent terms, and convexity of the exponential moment transfers both bounds to the average.

*Range-based alternative.* Hoeffding's inequality applied to each candidate separately and a union bound over the $m$ candidates give $|\widehat{u}_f - g(f)| \le r_1 = \sqrt{\frac{1}{2n} \ln \frac{2m}{\delta_1}}$ simultaneously with probability at least $1 - \delta_1$, so $B_{fb} = 2r_1$ for $f \neq b$ is also valid. Each family has level $\delta_1$ and their minimum has level $2\delta_1$. □

For classification the paired second moment $\widehat{q}_{fb}$ is the fraction of units on which the correctness indicators of the two models differ. Two candidates that make the same mistakes are therefore separated by far fewer units than the range-based radius requires, and in the extreme case of complete agreement the radius is at most $\frac{8L}{3n}$.

### 2.5 Anytime-valid paired confirmation

**Proposition 2.5** (Proposition 2 of the paper)**.** *Freeze an ordered family $f_{\mathrm{ch},1}, \dots, f_{\mathrm{ch},K}$ of challengers and the reference $f_{\mathrm{ref}}$ before the confirmation split is read. Let the units of $\mathcal{V}_{\mathrm{conf}}$ be independent draws from the task distribution, independent of everything the pair was built from, and let $z_1^{(j)}, z_2^{(j)}, \dots \in [-1, 1]$ be the per-unit improvements of $f_{\mathrm{ch},j}$ over $f_{\mathrm{ref}}$, read in an order under which $\mathbb{E}[z_t^{(j)} \mid \mathcal{H}_{t-1}] = \Delta_j$ for every $t$, where $\mathcal{H}_{t-1}$ is generated by the first $t-1$ units. Fix $\varepsilon \in [0, 1)$ and $\delta \in (0, 1)$, and define*

$$M_0^{(j)} = 1, \qquad M_t^{(j)} = M_{t-1}^{(j)} \left(1 + \lambda_t^{(j)} \left(z_t^{(j)} + \varepsilon\right)\right),$$

*where the stake $\lambda_t^{(j)} \in \left[0, \frac{1}{2(1-\varepsilon)}\right]$ is a function of $z_1^{(j)}, \dots, z_{t-1}^{(j)}$. The rule adopts the first challenger in the order with $\max_{t \le n} M_t^{(j)} \ge \frac{K}{\delta}$ and retains the reference when no challenger reaches the threshold. Then the probability that the rule adopts a challenger with $\Delta_j \le -\varepsilon$ is at most $\delta$, simultaneously for every $n$ and every stopping time.*

*Proof.* Fix $j$ with $\Delta_j \le -\varepsilon$ and drop the superscript. Each factor is nonnegative, because $z_t + \varepsilon \ge -(1 - \varepsilon)$ and $\lambda_t \le \frac{1}{2(1-\varepsilon)}$ give $\lambda_t(z_t + \varepsilon) \ge -\frac{1}{2}$. The stake is $\mathcal{H}_{t-1}$-measurable, so

$$\mathbb{E}\left[M_t \mid \mathcal{H}_{t-1}\right] = M_{t-1}\left(1 + \lambda_t(\Delta_j + \varepsilon)\right) \le M_{t-1},$$

and $(M_t)_{t \ge 0}$ is a nonnegative supermartingale with $M_0 = 1$. Ville's inequality [17] then gives

$$\Pr\left(\sup_t M_t \ge \frac{K}{\delta}\right) \le \frac{\delta}{K}.$$

The event that the rule adopts a challenger with $\Delta_j \le -\varepsilon$ is contained in the union of these events over the challengers with $\Delta_j \le -\varepsilon$, of which there are at most $K$, so a union bound gives $\delta$. The supremum over $t$ is what makes the statement hold at every stopping time, hence the split may be read incrementally and reading may stop at a data-dependent time. □

The deployed stake is the one-sided aGRAPA plug-in

$$\lambda_t = \min\left\{\frac{1}{2(1-\varepsilon)},\ \max\left\{0, \frac{\widehat{\mu}_{t-1}}{\widehat{\sigma}^2_{t-1} + \widehat{\mu}^2_{t-1}}\right\}\right\},$$

with $\lambda_1 = 0$ and with $\lambda_t = 0$ when $\widehat{\sigma}^2_{t-1} + \widehat{\mu}^2_{t-1} = 0$, computed from the running mean and variance of the shifted improvements [19]. The stake lies in the admissible interval and reads only the earlier units, so Proposition 2.5 applies to it. The next remark gives the number of confirmation units that a constant stake needs.

**Remark 2.6** (Remark 1 of the paper)**.** *Let $z_t \in [-1, 1]$ be independent and identically distributed with $\Delta = \mathbb{E}z_t > 0$ and $q = \mathbb{E}z_t^2$, fix $\varepsilon = 0$, and fix a constant stake $0 < \lambda \le \frac{1}{2}$ before the confirmation split is read. Put $\psi = \lambda\Delta - \lambda^2 q$, $a = \ln\frac{K}{\delta}$, and $\ell = \ln\frac{1}{\beta}$. With probability at least $1 - \beta$,*

$$\ln M_n \ge n\psi - 2\lambda\sqrt{nq\ell} - 2\lambda\ell. \tag{4}$$

*The right side being at least $a$ therefore guarantees that the wealth crosses the threshold by unit $n$ with probability at least $1 - \beta$. For correctness differences, $q = \Pr(z \ne 0) \ge \Delta$, and the admissible choice $\lambda = \frac{\Delta}{2q}$ crosses the threshold once*

$$n \ge \frac{q}{\Delta^2}\left(8\ln\frac{K}{\delta} + 24\ln\frac{1}{\beta}\right). \tag{5}$$

*Proof.* Write $Y_t = \ln(1 + \lambda z_t)$, so that $\ln M_n = \sum_{t\le n} Y_t$. For $u \in [-\frac{1}{2}, \frac{1}{2}]$ the inequality $\ln(1+u) \ge u - u^2$ holds, hence $\mathbb{E}Y_t \ge \lambda\Delta - \lambda^2 q = \psi$. On the same interval $|\ln(1+u)| \le \sqrt{2}\,|u|$, because the ratio is largest at $u = -\frac{1}{2}$, where it equals $2\ln 2 < \sqrt{2}$. Therefore $\mathbb{E}Y_t^2 \le 2\lambda^2 q$ and $\mathrm{Var}(Y_t) \le 2\lambda^2 q$. The range of $Y_t$ is $\ln\frac{1+\lambda}{1-\lambda} \le 3\lambda$ for $\lambda \le \frac{1}{2}$, so $Y_t - \mathbb{E}Y_t \ge -3\lambda$.

Bernstein's inequality for the lower tail, with variance proxy $\sigma^2 = 2\lambda^2 q$ and bound $3\lambda$, gives for every $t > 0$

$$\Pr\left(\sum_{t\le n} Y_t \le n\mathbb{E}Y_1 - t\right) \le \exp\left(-\frac{t^2}{2n\sigma^2 + 2\lambda t}\right).$$

Take $t = 2\lambda\sqrt{nq\ell} + 2\lambda\ell$. Then

$$t^2 = \lambda^2\left(4nq\ell + 8\ell\sqrt{nq\ell} + 4\ell^2\right), \qquad 2n\sigma^2 + 2\lambda t = \lambda^2\left(4nq + 4\sqrt{nq\ell} + 4\ell\right),$$

so $t^2 - \ell\left(2n\sigma^2 + 2\lambda t\right) = 4\lambda^2\ell\sqrt{nq\ell} \ge 0$ and the exponent is at most $-\ell$. The deviation probability is therefore at most $\beta$, and $\mathbb{E}Y_1 \ge \psi$ gives (4).

For (5), the choice $\lambda = \frac{\Delta}{2q}$ is admissible because $\Delta \le q$, and it gives $\psi = \frac{\Delta^2}{4q}$. Writing $n = \frac{q}{\Delta^2}(8a + 24\ell)$ yields $n\psi = 2a + 6\ell$ and

$$2\lambda\sqrt{nq\ell} = \frac{\Delta}{q}\sqrt{nq\ell} = \sqrt{(8a + 24\ell)\ell} \le a + 3\ell + 2\ell,$$

by $\sqrt{xy} \le \frac{x}{8} + 2y$ with $x = 8a + 24\ell$ and $y = \ell$, while $2\lambda\ell = \frac{\Delta}{q}\ell \le \ell$. The right side of (4) is then at least $2a + 6\ell - (a + 5\ell) - \ell = a$, and the threshold is crossed. Larger $n$ only increases the right side, since $\psi > 0$ under $\Delta > 0$ and $\lambda \le \frac{\Delta}{2q}$. □

# 3 Further formal results

## 3.1 Hardness of the record-level selection step

The manuscript states that the record-level search over the $\binom{N}{k}$ subsets of the pool contains the NP-hard maximum-coverage problem. The following statement makes that reduction explicit and records what it implies for the greedy solvers inside the strategy library.

**Proposition 3.1.** *Consider the decision form of record-level selection. Given a pool $\mathcal{D}$, unit costs, a budget $k$, a set utility $U$ evaluable in time polynomial in the input, and a threshold $t$, decide whether some $S \subseteq \mathcal{D}$ with $|S| \le k$ satisfies $U(S) \ge t$. This problem is NP-complete.*

*Proof.* Membership in NP follows from the witness $S$, whose cardinality is checked by counting and whose utility is checked by one evaluation of $U$, both in polynomial time. For hardness, instantiate $U$ as a coverage function. Each record $x$ carries a set $T(x)$ of ground elements, and $U(S) = |\bigcup_{x \in S} T(x)|$, which is evaluable in polynomial time. The decision form is then exactly budgeted maximum coverage with unit costs, which is NP-complete [9], and the instance map is the identity on this special case. □

Exact optimization of the record-level step is therefore out of reach unless P = NP, which is why every strategy of the library solves a surrogate. Greedy maximization attains the factor $1 - \frac{1}{e}$ for a nonnegative monotone submodular utility under a cardinality budget [15], and a partial-enumeration cost-benefit greedy preserves that factor under a token budget [18]. No polynomial algorithm attains $1 - \frac{1}{e} + \epsilon$ on the coverage special case unless P = NP [6]. The duplicate and outlier filters of a cooperative strategy make the composed objective non-monotone, so the inner selector is a heuristic for it, and the objective value cannot rank strategies. This is one reason the election reads validation utilities.

## 3.2 Bounded influence of contaminated validation

Finding 7 of the paper relabels or reweights part of the validation data and measures what the election loses. The following bound states what any such contamination can do to the statistics that Theorem 2.4 controls, and it holds against an adversary who sees the sample values and the portfolio.

**Proposition 3.2.** *Let the validation utility be the mean of per-unit scores in $[0, 1]$ over $n$ units drawn independently, and let at most a fraction $\rho_v$ of the units be modified arbitrarily after the draw. Then every single-strategy utility estimate shifts by at most $\rho_v$, every paired difference of estimates shifts by at most $2\rho_v$, and the constant 2 cannot be improved. Replacing $B_{fb}$ by $B_{fb} + 2\rho_v$ restores conclusions (a) and (b) of Theorem 2.4.*

*Proof.* Compare the contaminated sample with the counterfactual uncontaminated sample of the same units. A single-strategy mean has per-unit scores in $[0, 1]$, so modifying one unit moves it by at most $\frac{1}{n}$, and at most $\rho_v n$ units are modified, which gives the first bound. A paired difference has per-unit values in $[-1, 1]$, so modifying one unit moves its mean by at most $\frac{2}{n}$, which gives the second bound. The comparison is made unit by unit against a fixed counterfactual, so an adversary who chooses the modifications after seeing the sample gains nothing. Tightness follows by flipping $\rho_v n$ paired differences from $+1$ to $-1$, which moves the paired mean by exactly $2\rho_v$. The concentration event of Theorem 2.4 bounds the deviation of the uncontaminated paired difference, and the triangle inequality adds $2\rho_v$ to the radius, which preserves (a) and (b). □

### 3.3 Conjunctive composition of mechanism-matched detectors

A cooperative strategy gives each signal one role. The following lemma states the property that makes the composition preferable to averaging when each error mechanism has a matched detector.

**Lemma 3.3.** *Let the data errors fall into $H$ mechanisms, let mechanism $j$ have a matched detector whose quantile rank $r_j$ places every mechanism-$j$ record in the bottom quantile $q_j$, and let no detector rank clean records below uniform, that is $\Pr(r_j(x) \le s) \le s$ for every clean $x$ and every $s \in [0,1]$. Put $\bar{q} = \max_j q_j$. Then the conjunctive composite $r_\wedge(x) = \min_j r_j(x)$ removes every dirty record at threshold $\bar{q}$ while removing an expected clean fraction of at most $H\bar{q}$, and there are instances in which the conjunction removes a dirty record while the average of the ranks removes a clean record first.*

*Proof.* A mechanism-$j$ record satisfies $r_j \le q_j \le \bar{q}$, hence $r_\wedge \le \bar{q}$, so thresholding $r_\wedge$ at $\bar{q}$ removes every dirty record. A clean record is removed only if some detector ranks it in its bottom $\bar{q}$, an event of probability at most $\bar{q}$ for each detector, so a union bound over the $H$ detectors caps the expected removed clean fraction at $H\bar{q}$. For the separation, take a duplicate detector and a label detector, a near-duplicate with ranks $(0.05, 0.95)$, and a clean record with ranks $(0.30, 0.40)$. At threshold 0.10 the conjunction removes the near-duplicate, whose minimum is 0.05, and keeps the clean record, whose minimum is 0.30. The average ranks the clean record at 0.35 below the near-duplicate at 0.50, so a rule that removes the lowest average removes the clean record first. □

This is the formal counterpart of Example 2 of the paper. The admission step of a cooperative strategy carries the label-conflict role, the two filters carry the outlier and near-duplicate roles through neighbor distances, and no record is removed because an averaged score happened to be low.

### 3.4 Why one validation unit serves the whole portfolio

In a query model where each query consumes a fresh held-out unit and reveals the per-unit metric of a single candidate, identifying a near-optimal candidate costs a number of queries that grows linearly in the number of candidates. Adjudication does not operate in that model. One validation unit scores every frozen candidate, since the train-once cache holds one fit per distinct subset and each fit is scored on every split, so the union bound of Theorem 2.4 replaces the linear dependence on $m$ by $\ln m$, and Theorem 2.3 shows that this logarithmic dependence cannot be improved. The union bound runs over every candidate that the ranking split compares, grid cells included, so $m$ counts grid cells as candidates.

## 4 Method and protocol details

### 4.1 Features of the learned cleanliness score

The learned cleanliness score is a classifier of construction records against pool records, fitted on per-record features that correspond to the three quality signals. For classification tasks the features are the agreement of the record's label with the labels of its nearest pool neighbors, which carries authenticity, the probability and the margin of the record's label under a model cross-fitted on the pool, which carries influence, and the logarithms of the distances to the nearest pool neighbors, which carries coverage. Every feature is computed against the pool for pool records and construction records alike, so the two sources differ only in their position in the same feature space. The classifier weights the two sources equally, which is the equal-prior condition of Proposition 2.1, and every

pool score is cross-fitted, so no record is scored by a model that was fitted on it. The score reads the pool and the construction split only, which keeps it inside premise (P3).

### 4.2 Other subset objectives in the literature

Beyond per-record scores, two further families of subset objectives appear in the literature. One estimates the effect of removing a record on generalization and selects the records whose removal hurts most. The other trades per-record importance against pairwise similarity inside the selected set, which is the formulation that the MMR reranking rule [2] introduced for retrieval. Both families enter the frozen portfolio of OmniSelect in the same way as any other strategy, since a portfolio member is only required to return a subset within the budget.

### 4.3 Portfolio membership of external methods

Every method that the manuscript evaluates can enter the portfolio as one more candidate, so an election can return it on the tasks where it is the best candidate. The portfolio of the reported runs contains the library strategies, the fusion and cooperative grids, and the external baselines of each track. Besides the single-signal strategies, the library holds semantic deduplication, quality-diversity filtering, datamodel-based selection, farthest-first coverage, importance-similarity trade-offs, validation-guided subset selection, gradient matching, importance reweighting, gradient similarity, and compression, and its gradient-based members supply the gradient mechanism of the mechanism table. Each strategy enters the library on the tasks where its inputs exist. The three adaptations that the manuscript evaluates separately, namely EL2N, GraNd and CCS on forecasting and text, are excluded there because their definitions need labeled class probabilities, which those tracks do not provide. The membership of each track, with the reason for every exclusion, is recorded in the repository.

### 4.4 Robustness conditions

The deployment experiment varies the pool, the validation data, and the learner. The injection conditions I0 to I60 set the dirty share of the pool from 0 to 60 percent. Condition U replaces the error mechanisms by mechanisms that the authenticity detectors were not built for. Conditions L1 and L2 replace the downstream learner. Condition N30 relabels 30 percent of the validation examples at random, and the validation-noise panel extends this to half of them. Condition P50 halves the validation share of a random half of the classes, which shifts the class prior of the validation data while leaving the pool unchanged. The scale-up conditions enlarge the pool instead, to native ImageNet-100 and to the enlarged text pool.

## 5 Further experimental results

This section gives further measurements behind Findings 2, 5, and 6 of the paper. The Best baseline row of Table 4 of the paper takes, on each task, the external baseline with the best mean in Table 3 of the paper, which is DMF on C-100, C-100N, ETTh1, ETTm1, and TEP21, QuaDMix on ETTh2 and Steam, the CCS adaptation on CSTR, CCS on Electricity, and Herding on Text.

### 5.1 Election per task

Table 1 breaks the paired verdicts and the election of the paper down by task. The 69 baselines below Random are the per-task counts behind Finding 2, and the mean gains 12.5% and 15.1% of

the best reference and of the elected strategy are those of Finding 6.

Table 1: Paired verdicts and election on each task. W and T count the paired verdicts of OmniSelect against the baselines (it loses to none), Below the baselines whose mean falls below Random, $m$ the frozen candidates of a run, and Adopt the runs in which a challenger replaces the best reference. Kinds are fusion (Fu), cooperative (Co), consensus (Cs), library (Li), and external (Ex). The gains over Random (%) are those of the best reference (Ref.), the elected strategy, and the best frozen candidate in hindsight (Best).

| | Baselines | | | Election | | | Gain over Random (%) | | |
|---|---|---|---|---|---|---|---|---|---|
| Task | W | T | Below | $m$ | Adopt | Kinds | Ref. | Elected | Best |
| C-100 | 10 | 2 | 7 | 32 | some | Fu | 21.4 | 22.4 | 22.5 |
| C-100N | 12 | 0 | 6 | 32 | all | Co, Cs | 16.8 | 24.8 | 24.8 |
| C-10 | 11 | 1 | 7 | 22 | none | Ex, Li | 2.6 | 2.6 | 2.7 |
| ETTh1 | 12 | 0 | 3 | 29 | all | Fu, Co | 9.9 | 13.2 | 13.8 |
| ETTm1 | 12 | 0 | 3 | 29 | all | Co, Fu | 18.0 | 19.7 | 19.7 |
| ETTh2 | 12 | 0 | 8 | 29 | all | Fu, Cs | 5.2 | 12.0 | 12.7 |
| CSTR | 12 | 0 | 8 | 29 | none | Co | 7.1 | 7.1 | 8.0 |
| Steam | 12 | 0 | 8 | 29 | all | Fu | 23.8 | 27.7 | 27.9 |
| TEP21 | 11 | 1 | 2 | 33 | some | Li, Co, Fu | 30.5 | 32.9 | 33.4 |
| Elec | 12 | 0 | 7 | 34 | some | Co, Li | 1.6 | 2.9 | 2.9 |
| Text | 8 | 1 | 10 | 20 | none | Li | 0.6 | 0.6 | 0.6 |
| All | 124 | 5 | 69 | 20 to 34 | 64% | 5 | 12.5 | 15.1 | 15.4 |

### 5.2 Portfolio ceiling

A cooperative candidate holds the ceiling of the frozen portfolio, its best test utility in hindsight, in nearly a third of the runs.

## 6 Baseline fidelity

Each external baseline is implemented from its own selection rule under the shared protocol. Its inputs differ by method and by data type, and each entry below names them. This section records, for the two methods that fuse several signals, what the implementation retains and where it departs from the published formulation. The repository holds the same record next to the code, together with the fidelity tier of every other baseline.

### 6.1 QuaDMix

The symbols of this subsection are those of the published paper and are local to it. The implementation retains the structure of the published sampler. It merges normalized quality criteria inside a domain, converts the merged score into a within-domain percentile rank, and maps the rank through a sigmoid in the rank threshold $\omega$ with slope $\lambda$, so that records above the threshold keep only the floor $\epsilon$. It departs from the published formulation in the following respects.

- The sigmoid weight is implemented as $\frac{\eta}{1+\exp(-\lambda(\omega-r))}+\epsilon$ inside the threshold, while Equation (3) of the published paper uses $\left[\frac{2}{1+\exp(-\lambda(\omega-r))}\right]^{\eta}+\epsilon$. At the default $\eta=1$ the two differ by the

factor 2 on the sigmoid part, and the floor $\epsilon$ is not scaled with it, so the induced sampling distributions are close but not identical.

- The percentile rank is accumulated over records sorted by the merged score, so records with equal scores receive different ranks, while the published definition assigns the whole tie group the rank of its last element.
- Token weights default to one per record outside the text track, where the token counts of the documents are used.
- Domains are the native domain labels where a track provides them and $k$-means clusters of the representation otherwise.
- Expected repeat counts are converted into a fixed-size subset by Gumbel top-$k$ sampling without replacement, because the benchmark fixes the budget of every strategy, as a record count on the image, forecasting, process and tabular tasks and as a per-domain token count on the text tasks.
- The quality input is the authenticity signal outside the text track and the reference-model loss inside it. The published proxy-model training and the parameter search over the sampling parameters are not run, and the implementation uses fixed default parameters, which is why the mechanism table marks the method as reading no target-task validation feedback.

### 6.2 DMF

The implementation retains the fusion structure and the centered additive weight update of the published method. The actor scores are combined into one score by the published fusion form, and the weights are updated by the centered additive rule of the published update equation. It departs from the published formulation in the following respects.

- The actors are the three quality signals of OmniSelect, namely authenticity, influence and coverage, in place of the published quality, domain and topic actors.
- The reward of an actor is the construction-split utility of the subset that this actor alone selects, in place of the published aggregation of category weights and category rewards. The subset that the current fused weights select is scored as well, and that score only decides which iteration is reported.
- The actor-memory update of the published method is not run, and the actor set is fixed over the iterations.
- The updated weights are clipped at zero and renormalized, a step the published update equation does not contain.
- The reported subset is the best of the iterations under the construction-split score, rather than the subset of the last iteration.
- On the text track the reward is the negative mean per-domain loss of a reference set after a short fine-tuning run, which is the closest available counterpart of the published reward on that track.

### 6.3 Other adaptations

The remaining baselines are implemented from their published rules with the adaptations that a shared protocol requires. EL2N and GraNd score records by early-training error and by a last-layer gradient norm, and CCS allocates the budget across score strata. Their published scores need supervised class probabilities, so on forecasting and text the manuscript reports separately evaluated adaptations, marked /adapt, which are not members of the adjudicated portfolio of those runs. Herding matches the mean of the representation greedily and selects without replacement. Density samples by inverse neighbor density, with a $k$-nearest-neighbor estimator in place of the published locality-sensitive estimator. Each of these methods reports under the same pool, split, budget, learner and metric as every other row, so the comparison is a unified-budget comparison rather than a reproduction of the absolute numbers of the original papers. The repository records the fidelity tier and the evidence of each implementation.

## 7 Dataset provenance

Every task is built from a public source, and several tasks use a derived version of it. The table below gives the source and the derived version that the runs read. The repository holds the same list in `docs/dataset_provenance.md`, with the pinned revision or checksum of every downloaded artifact and the scripts that rebuild each pool.

| Task | Source | Derived version read by the runs |
|---|---|---|
| C-100, C-10 | CIFAR-100 and CIFAR-10 [11] | Official train and test splits |
| C-100N | CIFAR-100N [20] | Human label set over CIFAR-100 |
| IN-100 | ImageNet [4] | 100-class subset, images resized |
| ETTh1, ETTm1, ETTh2 | ETT streams [21] | Sliding windows of the released series |
| CSTR, Steam | DaISy records [14] | Released identification records |
| TEP21 | Tennessee Eastman process [5] | Public redistribution of the simulated runs |
| Elec | Electricity [7] | OpenML distribution of the dataset |
| Text | FineWeb-Edu [16], FineMath [8], CodeParrot [3], COCO captions [12], Adult [10] | Five-domain pool, documents concatenated to a common token length |

The image tasks read the official splits of their sources, and the ImageNet-100 pool is the 100-class subset of ImageNet with resized images, so the class list and the resizing are part of the derived version rather than of the source. The text pool concatenates documents of the five sources and keeps the test data of each domain disjoint from the pool, and the Adult records are rendered as text with one sentence per record. The forecasting tasks read the released series and convert them into windows, so the window length and the stride are part of the derived version. The process task reads one record per row of the simulated runs, with the process variables as its features and the normal class together with the fault classes as its labels.

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